(LVDTs). Encoders are digital sensors commonly used to provide position feedback for actuators. These consist of a glass or plastic disk that rotates between a light source (LED) and a pair of photo-detectors. Plastic disk is encoded with alternate light and dark sectors, so pulses are produced as the disk rotates.

LVDTs consist of a magnetic core that moves in a cylinder. These are commonly used for position feedback in servomechanisms, automated measurement, and many other industrial and scientific applications.

Force sensors. Force and pressure sensors usually act as transducers; these generate signals as a function of the pressure being imposed on them. Force-sensing resistors are two-pin force sensors whose resistance changes when a force, pressure or mechanical stress is applied on the sensor surface. FN3050 sensor from TE Connectivity is a rugged force load cell, highly suited for process industry and test bench applications.

Vibration/acceleration sensors. Ceramic piezoelectric sensors or accelerometers are most commonly used to detect vibration. Triaxial accelerometers are used in mobile systems, cars, turbines and aircrafts. These provide vibration information and position data. Inertial measurement units measure linear and angular motion usually with gyroscopes and accelerometers. These are widely used in aircraft and missile navigation and guidance.

Level sensors. These sensors are very common in industrial process control. Selection of a suitable level sensor depends on its size and geometry. Industrial process control includes hydrostatic and optical level sensors, ranging from simple limit-value detection to precision continuous level sensing. Hydrostatic sensors can be installed as submersible sensors for positioning in the fluid or with a screw thread for attachment to the exterior tank wall.

Fig. 2: Capacitive and inductive proximity sensors (Image courtesy: https://cdn.automationdirect.com)

Industrial wireless sensors

Most traditional sensors usually require a wire or cable to connect with the external instruments. The cable can be copper wire, twisted pair or fibre optic. However, transmission line cables cause signal and power losses. Also, wired network installation is cumbersome and takes much time. Wireless sensors offer flexibility of installation, resulting in improved process monitoring and control and also reduced installation and maintenance costs.

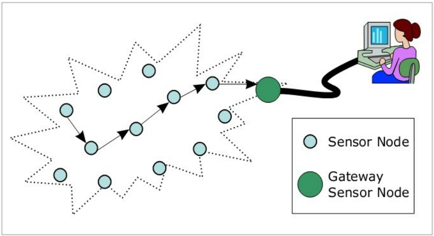

Fig. 3: Typical multi-hop wireless sensor network architecture (Courtesy: https://commons.wikimedia.org)

In industrial applications, wireless data transmission has been around for a long time. Monitoring temperatures, flow rates, levels and pressure parameters through wireless technology is becoming a popular trend in industrial applications. With the advent of Internet of Things (IoT), wireless sensor networks are deployed in some critical industrial applications. Wireless sensors used in supervisory control and data acquisition (SCADA) effectively address the needs of various industrial applications.

Many companies have been focusing on integrating miniature and low-power wireless radio with sensors. Wireless modules with pressure, flow, and temperature monitoring and control options are being developed to reduce installation and cable costs in large commercial and industrial buildings.

Recent advances in electronics have enabled the roll-out of miniature, low-power and multi-functional sensors that are suitable for short-distance industrial applications. Smart sensors connected through wireless link or wireless sensor network provide unprecedented opportunities for various industrial monitoring and control applications including the environment.

Wireless sensing of acceleration, vibration and velocity provides a more effective way of preventing equipment failures. Wireless sensors are not only wearable but also provide advantages of real-time, continuous, long-distance sensing and ease of operation.

There are many technologies used for wireless data transmission, including mobile phones, Zigbee, GPS, Wi-Fi,